RECONSTRUCTED ACADEMIC CV - Asım Orhan Barut

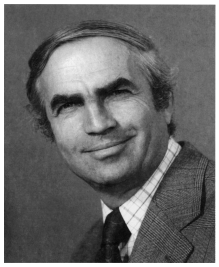
1926–1994 | Reconstructed from accepted source-linked records | E2K v75 FROZEN

Asım Orhan Barut

Academic curriculum vitae, reconstructed rather than an original historical document.

Portrait source: Inomata et al. (1993), unnumbered portrait preceding p. 173

Image asset preserved from the frozen manuscript.



RESEARCH PROFILE

Dynamical groups, group representations and symmetry · Particle models: leptons, hadrons, magnetic and stable-particle models · Quantum electrodynamics and self-field theory · Electron theory, zitterbewegung and classical spinning particles · S-matrix, Regge poles and scattering theory · Foundations of quantum mechanics · Path integrals, relativity and mathematical methods · Nonlinear field equations and solitons · Atomic and molecular physics, hydrogen atom, Rydberg states

ACADEMIC APPOINTMENTS

1944 Barut - eth [SPR-00102]

1948 Barut - eth_applphys [SPR-00103]

1953 Barut - chicago [SPR-00104]

Barut - reed [SPR-00105]

Barut - syracuse [SPR-00107]

1961 Barut - lrl [SPR-00108]

1962 Barut - colorado [SPR-00109]

Barut - cern [SPR-00110]

Barut - ictp [SPR-00111]

Barut - mpi [SPR-00112]

1950 Denel Fizik araştırma asistanı (research assistant in experimental physics) — Eidgenössische Technische Hochschule [SPR-00406]

1953 postdoctoral researcher in theoretical physics (Rockefeller fellow) — Chicago Üniversitesi [SPR-00407]

1954 faculty member (assistant professor per Özemre) — Reed College [SPR-00408]

1956 asistan profesör (assistant professor) — Syracuse Üniversitesi [SPR-00409]

1964 co-founder; regular visiting scientist (many summers and sabbaticals; own office) — Abdus Salam Uluslararası Teorik Fizik Merkezi [SPR-00413]

1962 scientific collaboration (first Turkish department Barut collaborated with) — Ankara Üniversitesi Fizik Bölümü [SPR-00415]

EDUCATION

1948 ETH diploma (experimental diploma work), 1948 person: Asım Orhan Barut; thesis: Experimental diploma work at ETH Zürich (untitled in sources); institution: Eidgenössische Technische Hochschule Zürich (ETH) [SPR-00562]

1951 Asım Barut doctorate in Switzerland, 1951 origin: Asım Orhan Barut [SPR-04820]

1952 Doctorate, ETH Zürich (1952) person: Asım Orhan Barut; thesis: Electronenoptisches und statistiches Verhalten der Gittervervielfacher; institution: Eidgenössische Technische Hochschule Zürich (ETH) [SPR-00427]

Barut graduated from Malatya Lisesi person: Asım Orhan Barut; institution: Malatya Lisesi [SPR-00122]

ETH doctorate: Master records include 1951 and 1952; the year is not harmonized across frozen records.

RESEARCH POSITIONS, VISITS AND FELLOWSHIPS

Additional appointments and visiting appointments, where documented, appear in the Master and the institutional history.

RESEARCH PROJECTS AND FUNDING

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n.d. Joint science centre initiative (Barut and Tomak) [SPR-00136]

1948-1953 Experimental research in Enis Baş's laboratory [SPR-00565]

TEACHING

1964 ICTP opening conference (Joly Oteli, Trieste, autumn 1964) [P-BARUT-A039]

1964 Latest developments in S-matrix theory of particle physics (ICTP opening conference, 1964) [P-BARUT-A040]

1966 Barut's talk at the Milwaukee conference on Non-compact Groups (1966) [P-BARUT-A041]

1966 Milwaukee conference on Non-compact Groups [P-BARUT-A042]

1969 50 bin bilim adamı için hamle [P-BARUT-A044]

1970-06-29 Introduction to De Sitter and Conformal Groups and Their Physical Applications (Boulder, 1970) [P-BARUT-A046]

1970-06-29 Representations of the Dynamical Group O(4,2) Realized in the Dyonium Atom (Boulder, 1970) [P-BARUT-A047]

n.d. Yaşam Öyküm (speech at the honor meeting organized for Barut) [P-BARUT-A062]

1980-03-06 İki Bileşenli Pomeron İçin Dinamik Bir Model (TFD-3 congress) [P-THE-BARUT-0032]

1987-09-16 Panel “Türkiye’de Bilim Adamı Yetiştirilmesi” (TFD-9 congress) [P-THE-BARUT-0033]

1990-09-19 Kuantumelektrodinamik, Kuantumoptik ve Kuantum Teorisi Temellerindeki Yeni Teorik ve Deneysel Sonuçlar (TFD-12 congress) [P-THE-BARUT-0034]

SUPERVISION

1971 K. Gediz Akdeniz — doctoral; unverified_formal_supervision; International Centre for Theoretical Physics (ICTP), Trieste | unresolved_supervision

1967 Hagen Kleinert — doctoral; formal_phd; University of Colorado

1972 İsmail Hakkı Duru — doctoral; formal_phd; University of Colorado

1988 Jonathan P. Dowling — doctoral; formal_phd; University of Colorado

1994 Ömer — doctoral; formal_phd; University of Colorado

1972 Haluk Beker — doctoral; formal_phd; None

Master's supervision, formal doctoral supervision, and qualified scientific mentorship are separate relation types; unverified roots are not promoted to confirmed doctoral supervision.

AWARDS AND HONORS

1982 Barut received the TÜBİTAK Bilim Ödülü (1982) [SPR-00131]

1982 Honorary doctorate from Karadeniz Teknik Üniversitesi (1982) [SPR-00440]

1987-05 Honorary doctorate from İnönü Üniversitesi (May 1987) [SPR-00443]

PROFESSIONAL SERVICE AND ADMINISTRATIVE ROLES

1960 Boulder Summer Schools on Theoretical Physics (University of Colorado) [P-BARUT-A008]

1960 Important scientific meetings in İstanbul in the 1960s and 1970s (organized by Erdal İnönü, Asım Barut and Feza Gürsey) [P-BARUT-A009]

1960 Scientific conference in Turkey organized by Asım Orhan Barut (early 1960s) [P-BARUT-A010]

1967 NATO Yaz Okulu, İstanbul, 1967 [P-BARUT-A013]

1970 NATO Yaz Okulu, İstanbul, 1970 [P-BARUT-A015]

1972 NATO Yaz Okulu, İstanbul, 1972 [P-BARUT-A018]

1979 UNESCO Matematiksel Fizik Kollokyumu, Trabzon, 1979 [P-BARUT-A020]

1981 NATO Yaz Okulu (İstanbul, 1981) [P-BARUT-A021]

1982 Yüksek Enerji Fiziği Araştırma Okulu (Trabzon, 1982) [P-BARUT-A022]

1983 NATO Yaz Okulu (İstanbul, 1983) [P-BARUT-A023]

1984 NATO Yaz Okulu (İstanbul, 1984) [P-BARUT-A025]

1989 NATO Yaz Okulu (İstanbul, 1989) [P-BARUT-A027]

1991-12 Uluslararası Matematiksel Fizik Konferansı (Edirne, December 1991) [P-BARUT-A028]

1993-12 Uluslararası Matematiksel Fizik Konferansı (Edirne, December 1993) [P-BARUT-A030]

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1994-09 Electron Theory and Quantum Electrodynamics: 100 Years Later [P-BARUT-A032]

n.d. International conference being prepared by A. O. Barut (unnamed, 1990s) [P-BARUT-A033]

n.d. NATO Advanced Study Institutes organized by Barut over his career (collective mention) [P-BARUT-A036]

n.d. NATO conference in Trabzon organized by A. O. Barut [P-BARUT-A037]

NAMED LEGACY (NOT AWARDS RECEIVED BY THE ORIGIN)

Bust of Asım Orhan Barut in front of the Edirne Karaağaç station building — [OBJ-00018]

Bronze bust of Asım Orhan Barut — [OBJ-00185]

BARUT EBRUSU — [OBJ-00186]

Asım Orhan Barut biyografi sitesi (asimorhanbarut.com) — [OBJ-00187]

Barut Memorial Lecture — [OBJ-02161]

Bust of Asım Orhan Barut, Department of Physics, Faculty of Science, Trakya University — Department of Physics, Faculty of Science, Trakya University [E-BARUT-L-001]

Barut's personal book collection (made available to students and researchers) — [E-BARUT-L-002]

Barut Foundation — Boğaziçi University lecture series (venue; not Foundation premises) [OBJ-02158]

PUBLICATIONS (589)

The accepted Module P catalogue is reproduced below. A record in this list is not necessarily a claim about the definitive lifetime publication total.

1926 GENERATION EVIDENCE ATLAS
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1. **1951** Die Dichte und das Periodische System der Elemente *Mitteilungen der G.F.F. (Zürich)* [P-BARUT-W0521]

2. **1951** Die Laufzeit, Elektronenbahnen, Kathodenfeldstärke und Potential der Raumladungsdiode für jede Anfangsgeschwindigkeit, Anfangsrichtung und Strom *ZAMP Zeitschrift für angewandte Mathematik und Physik* [P-BARUT-W0003]

3. **1951** The Cathode Field in Diodes under Partial Space-Charge Conditions with Initial Velocities *Physical Review* [P-BARUT-W0005]

4. **1954** The Mechanism of Secondary Electron Emission *Physical Review* [P-BARUT-W0008]

5. **1956** Anisotropy of cosmic rays due to galactic rotation *Il Nuovo Cimento* [P-BARUT-W0009]

6. **1956** Foundations of Quantum Statistics *Report on a Symposium on Semiconductors, National Research Council of Canada* [P-BARUT-W0523]

7. **1957** Distribution Functions for Noncommuting Operators *Physical Review* [P-BARUT-W0010]

8. **1958** A theory of particles of spin one-half *Annals of Physics* [P-BARUT-W0011]

9. **1958** Covariant Quantum Statistics of Fields *Physical Review* [P-BARUT-W0012]

10. **1958** On the symmetry of elementary particles *Il Nuovo Cimento* [P-BARUT-W0013]

11. **1959** Relativistic invariance and quantum mechanics *Il Nuovo Cimento (Supplemento)* [P-BARUT-W0014]

12. **1959** Vacuum expectation values and analytic functions *Il Nuovo Cimento (Supplemento)* [P-BARUT-W0015]

13. **1960** Extra solutions of the dispersion relations and resonance scattering *Nuclear Physics* [P-BARUT-W0016]

14. **1960** Spin-Orbit Correlations in μ−e and e−−e− Scattering *Physical Review* [P-BARUT-W0017]

15. **1960** The Kemmer β-formalism for particles of spin one-half *Il Nuovo Cimento* [P-BARUT-W0018]

16. **1961** Analytical Properties of S Matrix and Uniqueness of the Scattering Potential *Journal of Mathematical Physics* [P-BARUT-W0019]

17. **1961** Boson Currents in the Theory of Weak Interactions *Physical Review* [P-BARUT-W0524]

18. **1961** Kinematical and Dynamical Resonances *Physical Review* [P-BARUT-W0020]

19. **1961** Note on Gauge Transformations in Quantum Mechanics *American Journal of Physics* [P-BARUT-W0021]

20. **1962** Action principle for higher order Lagrangians with an indefinite metric *Annals of Physics* [P-BARUT-W0022]

21. **1962** Analyticity in Angular Momentum of the Relativistic Many-Channel S Matrix from Dispersion Relations and Unitarity *Physical Review* [P-BARUT-W0023]

22. **1962** Complex Angular Momentum in Relativistic S-Matrix Theory *Physical Review* [P-BARUT-W0024]

23. **1962** Dispersion Relations and Resonance Scattering *Lectures in Theoretical Physics, Vol. IV; ed. W. E. Brittin* [P-BARUT-W0525]

24. **1962** Formulation of the Scattering Functions in Terms of the Unitary Representations of the Inhomogeneous Lorentz Group *Physical Review* [P-BARUT-W0025]

25. **1962** Nature of the resonance in the static pion-nucleon scattering in one-meson approximation *Nuclear Physics* [P-BARUT-W0026]

26. **1962** QUANTUM FIELD THEORY AND ELEMENTARY PARTICLES [P-BARUT-W0028]

27. **1962** Quantization of two-component higher order spinor equations *Annals of Physics* [P-BARUT-W0027]

28. **1962** Singularities in Angular Momentum of the Scattering Amplitude for a Class of Soluble Potentials *Physical Review* [P-BARUT-W0029]

29. **1962** Virtual Particles *Physical Review* [P-BARUT-W0030]

30. **1963** Amplitudes and crossing relations in isospin *Il Nuovo Cimento* [P-BARUT-W0031]

31. **1963** Behavior of the Scattering Amplitude for Large Angular Momentum *Journal of Mathematical Physics* [P-BARUT-W0032]

32. **1963** Construction of Invariant Scattering Amplitudes for Arbitrary Spins and Analytic Continuation in Total Angular Momentum *Physical Review* [P-BARUT-W0033]

33. **1963** Crossing Symmetry in S-Matrix Theory *Physical Review* [P-BARUT-W0034]

34. **1963** Helicity Amplitudes and Complex Angular Momentum in V-A Interactions and Under γ5-Invariance *Proceedings of the Conference on Photon Interactions, MIT* [P-BARUT-W0526]

35. **1963** On the symmetry of elementary particles — II *Il Nuovo Cimento* [P-BARUT-W0035]

36. **1963** Recent Developments in S-Matrix Theory of Particles and Resonances *Lectures in Theoretical Physics, Vol. VI; ed. W. E. Brittin* [P-BARUT-W0527]

37. **1964** A symmetry group containing both the lorentz group and SU 3 *Il Nuovo Cimento* [P-BARUT-W0036]

38. **1964** Complex Lorentz Group with a Real Metric: Group Structure *Journal of Mathematical Physics* [P-BARUT-W0037]

39. **1964** Connection Between the Relativistic and Nonrelativistic Form of the Nucleon-Nucleon Amplitude *Physical Review* [P-BARUT-W0038]

40. **1964** Dynamical Symmetry Group Based on Dirac Equation and Its Generalization to Elementary Particles *Physical Review* [P-BARUT-W0039]

41. **1964** Dynamical Symmetry Groups and Mass Spectrum for Elementary Particles *Conference on Symmetry at Higher Energy* [P-BARUT-W0529]

42. **1964** Electrodynamics and Classical Theory of Fields and Particles *Macmillan* [P-BARUT-W0040]

43. **1964** I=0 Pion-Pion Amplitude and σ Di-Pion *Physical Review Letters* [P-BARUT-W0041]

44. **1964** Nucleon and the Pion-Nucleon Trajectory. Determination of the Pion-Nucleon Coupling Constant *Physical Review* [P-BARUT-W0042]

45. **1964** On the coupling constants of massless particles *Physics Letters* [P-BARUT-W0043]

46. **1964** Secondary Diffraction Peaks in πN and KN Scatterings *Physical Review Letters* [P-BARUT-W0047]

47. **1964** The Framework of S-Matrix Theory *Scottish Universities Summer Institute* [P-BARUT-W0528]

48. **1964** The exact determination of coupling constants in terms of masses from unitarity and analyticity *Physics Letters* [P-BARUT-W0048]

49. **1965** Analycity, complex and quaternionic Lorentz groups and internal quantum numbers *Lectures in Theoretical Physics, Vol. VII A* [P-BARUT-W0049]

50. **1965** Analytic properties of the partial wave amplitudes for production processes in the complex j-plane *Lectures in Theoretical Physics, Vol. VII A* [P-BARUT-W0050]

51. **1965** Dynamical Groups and Mass Formula *Physical Review* [P-BARUT-W0051]

52. **1965** Dynamics of a Broken SU_N Symmetry for the Oscillator *Physical Review* [P-BARUT-W0052]

53. **1965** Invariant Amplitudes and Interpretation of SU3 and SU3 Type Symmetries *International Seminar on High Energy Physics and Elementary Particles* [P-BARUT-W0530]

54. **1965** Lorentz Group. Proceedings, 7th Summer Institute for Theoretical Physics *Lectures in Theoretical Physics, Vol. VII A* [P-BARUT-W0053]

55. **1965** Mass Difference of Ξ− and Ξ0 *Physical Review* [P-BARUT-W0054]

56. **1965** Mass Spectrum from Non-Compact Groups *International Seminar on High Energy Physics* [P-BARUT-W0531]

57. **1965** On non-compact groups I. Classification of non-compact real simple Lie groups *Proceedings of the Royal Society of London. Series A* [P-BARUT-W0055]

58. **1965** On non-compact groups. II. Representations of the 2+1 Lorentz group *Proceedings of the Royal Society of London. Series A* [P-BARUT-W0056]

59. **1965** On the Position of Singularities of the S-Matrix in Unified Theories [P-BARUT-W0057]

60. **1965** Production Amplitudes. I. Construction of Invariant Amplitudes *Physical Review* [P-BARUT-W0058]

61. **1965** Production Amplitudes. II. Partial-Wave Decomposition and Analytic Continuation in Total Angular Momentum *Physical Review* [P-BARUT-W0059]

62. **1965** Proton-neutron magnetic moment ratio in Wigner's theory of supermultiplets *Physics Letters* [P-BARUT-W0060]

63. **1965** Relation between the coupling constant and the singularity in the second sheet — II *Il Nuovo Cimento* [P-BARUT-W0061]

64. **1965** Remarks on the group structure of corben's wave equations *Lectures in Theoretical Physics, Vol. VII B* [P-BARUT-W0062]

65. **1965** Renormalized perturbation series of electrodynamics from S-matrix theory *Il Nuovo Cimento* [P-BARUT-W0063]

66. **1965** S-Matrix Theory of Electromagnetic Interactions *Acta Physica Austriaca, Supplementum II* [P-BARUT-W0064]

67. **1965** Simultaneous analytic continuation to the second sheet in two crossed channels *Physics Letters* [P-BARUT-W0065]

68. **1966** Asymptotic Behavior of the Scattering Amplitude in the Left Half-Plane *Journal of Mathematical Physics* [P-BARUT-W0067]

69. **1966** Consequences of Three Discrete Internal Symmetry Transformations for Leptons *Physical Review* [P-BARUT-W0068]

70. **1966** Lectures in Theoretical Physics, Vol. VIII B *University of Colorado Press* [P-BARUT-W0514]

71. **1966** Near Forward Peaks in the K−p and π−p Charge-Exchange Scattering *Physical Review Letters* [P-BARUT-W0071]

72. **1966** On Irreducible Representation of a Class of Algebras and Related Projective Representations *Journal of Mathematical Physics* [P-BARUT-W0072]

73. **1966** The Two Uses of Noncompact Groups in Particle Physics and Their Relations *Noncompact Groups in Particle Physics* [P-BARUT-W0532]

74. **1966** Two examples of covariant theories with internal symmetries involving spin *Proceedings of the Royal Society of London. Series A* [P-BARUT-W0074]

75. **1966** Unitary and Antiunitary Ray Representations of the Product of n Commuting Parity Operators *Journal of Mathematical Physics* [P-BARUT-W0075]

76. **1967** Calculation of Relativistic Transition Probabilities and Form Factors from Noncompact Groups *Physical Review* [P-BARUT-W0076]

77. **1967** Calculation of Transition Probabilities from Noncompact Dynamical Groups *Physical Review* [P-BARUT-W0535]

78. **1967** Current Operators and Majorana Equation for the Hydrogen Atom from Dynamical Groups *Physical Review* [P-BARUT-W0077]

79. **1967** Dynamical Group O(4, 2) for Baryons and the Behavior of Form Factors *Physical Review* [P-BARUT-W0078]

80. **1967** Dynamical Groups *Proceedings of the NATO Advanced Study Institute, İstanbul (1966)* [P-BARUT-W0536]

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81. **1967** Dynamical groups and a criterion of elementarity *Lectures in Theoretical Physics* [P-BARUT-W0079]

82. **1967** Hadron properties from non-compact dynamical groups *Conference proceedings (INSPIRE conference code C670828)* [P-BARUT-W0080]

83. **1967** High energy and particle physics. Proceedings, 9th Summer Institute for Theoretical Physics *Lectures in Theoretical Physics, Vol. IX B* [P-BARUT-W0081]

84. **1967** Mathematical methods of theoretical physics. Proceedings, 9th Summer Institute for Theoretical Physics *Lectures in Theoretical Physics, Vol. IX A* [P-BARUT-W0082]

85. **1967** Meson Decay Rates in O(4, 2) and the Introduction of SU(3) *Physical Review Letters* [P-BARUT-W0083]

86. **1967** On the superconvergence of amplitudes involving massless particles *Nuclear Physics B* [P-BARUT-W0084]

87. **1967** Resonance Decays from O(3,1)-Dynamics. A Regularity in the Partial Decay Widths *Physical Review Letters* [P-BARUT-W0537]

88. **1967** Strong Decay of High Spin Baryon Resonances in O(4,2) Theory *Physical Review Letters* [P-BARUT-W0540]

89. **1967** Tensor Operators and Mass Formula in the Minimal Extension of U(3) by Charge Conjugation *Journal of Mathematical Physics* [P-BARUT-W0534]

90. **1967** The Solution of the Discrete Mass Problem with Internal Degrees of Freedom and Further Developments *Symmetry Principles at High Energies, Fourth Coral Gables Conference* [P-BARUT-W0538]

91. **1967** Theory of the scattering matrix for the interactions of fundamental particles *Macmillan* [P-BARUT-W0085]

92. **1967** Transition Form Factors in Hydrogen Atom *Physical Review* [P-BARUT-W0539]

93. **1967** Transition Probabilities of the Hydrogen Atom from Noncompact Dynamical Groups *Physical Review* [P-BARUT-W0086]

94. **1967** Unified algebraic construction of representations of compact and non-compact Lie algebras and Lie groups *Lectures in Theoretical Physics* [P-BARUT-W0087]

95. **1968** Application of the dynamical group theory to the structure of the hadrons *Lectures in Theoretical Physics* [P-BARUT-W0088]

96. **1968** Derivation of Mass Spectrum and Magnetic Moments from Current Conservation in Relativistic O(3, 2) and O(4, 2) Theories *Physical Review* [P-BARUT-W0089]

97. **1968** Evidence for O(4, 2), assignment of hadron levels *Physics Letters B* [P-BARUT-W0090]

98. **1968** High energy physics and fundamental particles. Proceedings, 10th Summer Institute for Theoretical Physics *Lectures in Theoretical Physics, Vol. X B* [P-BARUT-W0091]

99. **1968** Implications of the o(4.2) model of strong interactions for regge pole and high-energy phenomenology *Conference proceedings (INSPIRE conference code C680906)* [P-BARUT-W0092]

100. **1968** Magnetic Moments, Form Factors, and Mass Spectrum of Baryons *Physical Review Letters* [P-BARUT-W0093]

101. **1968** Matrix Elements of Representations of Noncompact Groups in a Continuous Basis *Communications in Mathematical Physics* [P-BARUT-W0543]

102. **1968** Pion form factor and mass spectrum of bosons *Nuclear Physics B* [P-BARUT-W0094]

103. **1968** Position Operators and Localizability of Quantum Systems Described by Finite and Infinite Component Wave Equations *Reviews of Modern Physics* [P-BARUT-W0541]

104. **1968** Proton-Proton Scattering and Nuclear Structure. The Scalar Interaction in O(4,2) *Physical Review* [P-BARUT-W0542]

105. **1968** Quantum theory and statistical physics. Proceedings, 10th Summer Institute for Theoretical Physics *Lectures in Theoretical Physics, Vol. X A* [P-BARUT-W0095]

106. **1968** Radiative decay of boson resonances *Nuclear Physics B* [P-BARUT-W0096]

107. **1968** Reformulation of the Dirac Theory of the Electron *Physical Review Letters* [P-BARUT-W0097]

108. **1968** The *anomalous zitterbewegung* of composite particles *Il Nuovo Cimento A* [P-BARUT-W0098]

109. **1969** Calculation of the k-l-3 form-factors *Physical Review* [P-BARUT-W0099]

110. **1969** Dynamical groups and their currents - a model for strong interactions *Springer Tracts in Modern Physics* [P-BARUT-W0100]

111. **1969** Elementary particle physics. Proceedings, 11th Summer Institute for Theoretical Physics, Part. 1: 5th Boulder Conference on Particle Physics, Pt. 1 -2 *Lectures in Theoretical Physics, Vol. XI A* [P-BARUT-W0101]

112. **1969** Form-factor and mass corrections to weak decay amplitudes *Nuclear Physics B* [P-BARUT-W0102]

113. **1969** Massless particles and analytic continuation in mass *Acta Physica Austriaca, Supplementum* [P-BARUT-W0103]

114. **1969** On the radiative corrections to the electromagnetic properties of hadrons *Il Nuovo Cimento A* [P-BARUT-W0104]

115. **1969** Pion-gauge conditions for electro- and photoproduction of n*(j=1/2) and equality of the vector and axial vector form factors *Nuclear Physics B* [P-BARUT-W0105]

116. **1969** Relativistic Infinite Component Wave Equation for H-Atom with Spin *Physics Letters A* [P-BARUT-W0547]

117. **1969** Scattering theory ; new methods and problems in atomic, nuclear, and particle physics [P-BARUT-W0106]

118. **1969** The Analytic S-Matrix and the Algebraic Theory of Scattering *Scattering Theory, ed. A. O. Barut* [P-BARUT-W0545]

119. **1969** The group-theoretical derivation of the pion gauge condition *Lettere al Nuovo Cimento* [P-BARUT-W0107]

120. **1969** The physical interpretation of spacelike and ghostlike solutions of infinite-multiplet wave equations *Lettere al Nuovo Cimento* [P-BARUT-W0108]

121. **1969** Theory of the heavy electron *Physical Review* [P-BARUT-W0109]

122. **1969** Theory of the relativistic h atom and positronium *Physical Review* [P-BARUT-W0110]

123. **1969** 'Self-Consistency' Condition for an Inelastic Amplitude and for Coupling Constants *Lettere al Nuovo Cimento* [P-BARUT-W0546]

124. **1970** A unified treatment of leptons *Il Nuovo Cimento A* [P-BARUT-W0111]

125. **1970** Crossing symmetry in the o(4,2) formulation of the dirac theory *Physical Review D* [P-BARUT-W0112]

126. **1970** On s-matrix models for weak interactions *Lettere al Nuovo Cimento* [P-BARUT-W0114]

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127. **1970** On the Electromagnetic Origin of Strong Interactions *Proceedings of the Coral Gables Conference on Fundamental Interactions at High Energies, II* [P-BARUT-W0548]

128. **1970** On the s-matrix theory of weak interactions *Springer Tracts in Modern Physics* [P-BARUT-W0115]

129. **1970** Realization of current-algebra commutation relations on the space of solutions of finite- or infinite-component wave equations *Physical Review D* [P-BARUT-W0116]

130. **1970** Reduction of a class of o(4,2) representations with respect to so(4,1) and so(3,2) *Journal of Mathematical Physics* [P-BARUT-W0117]

131. **1970** Simply factorizable n-point amplitude *Physical Review D* [P-BARUT-W0118]

132. **1970** Very-high-energy limit of proton proton scattering *Physical Review D* [P-BARUT-W0119]

133. **1970** Why is n*(1470) not observed in photoproduction experiments? *Lettere al Nuovo Cimento* [P-BARUT-W0120]

134. **1971** Atoms with Magnetic Charges as Models of Hadrons *Topics in Modern Physics – A Tribute to E. U. Condon* [P-BARUT-W0549]

135. **1971** De Sitter and Conformal Groups and their Applications. Proceedings, 13th Summer Institute for Theoretical Physics *Lectures in Theoretical Physics, Vol. XIII* [P-BARUT-W0121]

136. **1971** Die Theorie der Streumatrix für die Wechselwirkungen fundamentaler Teilchen *BI-Hochschultaschenbücher 438/438a* [P-BARUT-W0122]

137. **1971** New 'coherent' states associated with noncompact groups *Communications in Mathematical Physics* [P-BARUT-W0124]

138. **1971** Nonrelativistic and relativistic coulomb amplitude as the matrix element of a rotation in o(4,2) *Physical Review D* [P-BARUT-W0125]

139. **1971** Proton form-factor, magnetic charges, and dyonium *Physical Review D* [P-BARUT-W0126]

140. **1971** Representations of the dynamical group O(4,2) realized in the dyonium atom *Lectures in Theoretical Physics* [P-BARUT-W0127]

141. **1971** S0(4,2)-formulation of the symmetry breaking in relativistic Kepler problems with or without magnetic charges *Journal of Mathematical Physics* [P-BARUT-W0128]

142. **1971** The Relativistic Bohr Formula *Nature* [P-BARUT-W0550]

143. **1971** The nature of the nuclear bond and the structure of hadrons [P-BARUT-W0129]

144. **1972** A possible physical basis for some mathematical models of strong interactions: atoms with magnetic charges *Lectures in Theoretical Physics* [P-BARUT-W0130]

145. **1972** Bound state poles of the form-factors in eikonal approximation and infinite component wave equations *Nuclear Physics B* [P-BARUT-W0131]

146. **1972** Coupling of three or more positroniums - a relativistic model of coupling of composite particles *Nuclear Physics B* [P-BARUT-W0132]

147. **1972** Criterion for and the physical interpretation of minimal and nonminimal interactions *Annals of Physics* [P-BARUT-W0133]

148. **1972** Dynamical groups and generalized symmetries in quantum theory : with applications in atomic and particle physics *Canterbury Üniversitesi* [P-BARUT-W0134]

149. **1972** Infinite-component wave equations and mass splittings within su(3) and su(2) multiplets *Physical Review D* [P-BARUT-W0135]

150. **1972** On the Group Structure of the Periodic System of Elements *Proceedings Rutherford Centenary Symposium* [P-BARUT-W0551]

151. **1972** Properties of non-unitary zero mass induced representations of the Poincaré group on the space of tensor-valued functions *Annales de l'Institut Henri Poincaré A* [P-BARUT-W0136]

152. **1972** Review of hadron symmetries: multiplets, supermultiplets and infinite multiplets [P-BARUT-W0137]

153. **1972** Review of lepton multiplets and the composite structure of the muon [P-BARUT-W0138]

154. **1972** S-matrix approach to weak interactions *Fortschritte der Physik* [P-BARUT-W0139]

155. **1972** Scattering of Electromagnetic Radiation in Terms of Functions over the Group SU(2) *Aerospace Research Laboratories* [P-BARUT-W0552]

156. **1972** Superselection rule for magnetic charges and the search for magnetic monopoles *Physics Letters B* [P-BARUT-W0140]

157. **1972** The gravitational vertex in eikonal approximation *Lettere al Nuovo Cimento* [P-BARUT-W0141]

158. **1972** The infinite-dimensional lie algebra common to dual models, current algebras, dynamical groups and gauge groups *Lettere al Nuovo Cimento* [P-BARUT-W0142]

159. **1972** Theory of the conformally invariant mass *Annals of Physics* [P-BARUT-W0143]

160. **1972** Topics in Strong Interactions. Proceedings, 14th Summer Institute for Theoretical Physics *Lectures in Theoretical Physics, Vol. XIV A* [P-BARUT-W0144]

161. **1972** Value of the xi Parameter in Kl3 Decay from the Pion-Gauge Condition *Physical Review D* [P-BARUT-W0145]

162. **1973** A New Relativistic Coulomb Hamiltonian with O(4) Symmetry and a Spinor Realization of the Dynamical Group O(4,2) *Physical Review D* [P-BARUT-W0146]

163. **1973** Algebraic Solution of the Schrödinger Equation for a Class of Velocity-Dependent Potentials *Lettere al Nuovo Cimento* [P-BARUT-W0147]

164. **1973** Cancellations of Divergences in Fourth-Order μ-μ Elastic Scattering in the Gauge Model of Weak Interactions *Lettere al Nuovo Cimento* [P-BARUT-W0148]

165. **1973** Conformal Group–Schrödinger Group–Dynamical Group *Helvetica Physica Acta* [P-BARUT-W0557]

166. **1973** Conformally Invariant Massive Spinor Equations, I *Il Nuovo Cimento A* [P-BARUT-W0150]

167. **1973** Conformally Invariant Massive and Massless Spinor Equations, II. Solutions of the Wave Equations *Il Nuovo Cimento A* [P-BARUT-W0149]

168. **1973** Dispersion-Theoretic Calculation of the Lepton Magnetic Moment and Form Factors in the Gauge Theory of Weak Interactions *Nuclear Physics B* [P-BARUT-W0151]

169. **1973** External (Kinematical) and Internal (Dynamical) Conformal Symmetry, Discrete Mass Spectrum *2nd International Colloquium on Group Theoretical Methods* [P-BARUT-W0559]

170. **1973** Gauge Coordinates *2nd International Colloquium on Group Theoretical Methods* [P-BARUT-W0558]

171. **1973** H-Atom as a Relativistic Elementary Particle, I. Wave Equations and Mass Formula *Journal of Physics B* [P-BARUT-W0555]

1926 GENERATION EVIDENCE ATLAS
RECONSTRUCTED - PROJECT DOCUMENT
NOT AN ORIGINAL HISTORICAL DOCUMENT

172. **1973** H-Atom as a Relativistic Elementary Particle, II. Relativistic Coulomb Scattering and Relativistic Photoeffect *Journal of Physics B* [P-BARUT-W0556]

173. **1973** Improved Calculation of the Structure Functions of the Nucleon in the O(4,2) Infinite Multiplet Model *Physics Letters B* [P-BARUT-W0152]

174. **1973** Kℓ3 and πe3 Decays in Terms of the Kemmer β-Current *Physical Review D* [P-BARUT-W0153]

175. **1973** Magnetic Charges and Strings as Building Blocks of Hadrons: Connection between Electromagnetic and Dual Strings *Acta Physica Austriaca, Supplementum* [P-BARUT-W0154]

176. **1973** On Conformally Invariant Spinor Field Equations *Canadian Journal of Physics* [P-BARUT-W0155]

177. **1973** On Relativistic Models with Multiplet Structure *Lettere al Nuovo Cimento* [P-BARUT-W0156]

178. **1973** Scattering of Electromagnetic Radiation in Terms of Functions Over the Group SU(2) *Annals of Physics* [P-BARUT-W0554]

179. **1973** Solutions of the Conformally Invariant Spinor Equations and the Theory of the Electron-Muon System *Lettere al Nuovo Cimento* [P-BARUT-W0157]

180. **1973** Studies in Mathematical Physics. Lectures Presented at the NATO Advanced Study Institute, Istanbul, August 1970 [P-BARUT-W0158]

181. **1973** The Impossibility of Splitting the Superposition of States of Opposite Magnetic Charges in External Electromagnetic Fields: An Anti-Superselection Rule *Physics Letters B* [P-BARUT-W0159]

182. **1973** The Introduction of Internal Coordinates into the Infinite-Component Majorana Wave Equations *Proceedings of the Royal Society A* [P-BARUT-W0553]

183. **1973** The New Mathematical and Physical Problems of Superstrong Coupling of Magnetic Monopoles and Composite Hadrons *Studies in Mathematical Physics (lectures, NATO Advanced Study Institute, Istanbul, August 1970)* [P-BARUT-W0160]

184. **1973** Theory of Nucleon-Nucleon Scattering *Fortschritte der Physik* [P-BARUT-W0161]

185. **1974** Classical Relativistic Scattering of Two Dyons *Il Nuovo Cimento A* [P-BARUT-W0162]

186. **1974** Electrodynamics in Terms of Retarded Fields *Physical Review D* [P-BARUT-W0163]

187. **1974** Extended Singularities of the Electromagnetic Field as Models of Elementary Particles *Quantum Theory and the Structures of Time and Space (Feldafing 1974 proceedings)* [P-BARUT-W0164]

188. **1974** Group Theory in Nonlinear Problems. Lectures Presented at the NATO Advanced Study Institute on Mathematical Physics Held in Istanbul, August 7-18, 1972 [P-BARUT-W0165]

189. **1974** Relativistic Composite Systems and Extensible Models of Fields and Particles *Mathematical Physics and Physical Mathematics (Warsaw 1974 proceedings)* [P-BARUT-W0166]

190. **1974** Relativistic Composite Systems and Minimal Coupling *Physical Review D* [P-BARUT-W0167]

191. **1974** Relativistic Transition Amplitudes in a Class of O(4,2) Infinite Multiplets. I. Continuum-Continuum Transitions *Physical Review D* [P-BARUT-W0168]

192. **1974** Relativistic Transition Amplitudes in a Class of O(4,2) Infinite Multiplets. II. Discrete-Continuum Transitions *Physical Review D* [P-BARUT-W0169]

193. **1974** Spin-Statistics Connection for Dyonium *Physical Review D* [P-BARUT-W0170]

194. **1974** The Electromagnetic String with Spin *Nuclear Physics B* [P-BARUT-W0171]

195. **1974** Theory of the Electromagnetic Structure Functions of the Proton *Physical Review D* [P-BARUT-W0172]

196. **1974** Unification of the External Conformal Symmetry Group and the Internal Conformal Dynamical Group *Journal of Mathematical Physics* [P-BARUT-W0173]

197. **1975** Dilatation of Nonrelativistic Wave Functions into Relativistic Form [P-BARUT-W0174]

198. **1975** Electrodynamics in Terms of Functions over the Group SU(2), I. The Equation for Vector Potential *Foundations of Physics* [P-BARUT-W0560]

199. **1975** On the Measurability of Long-Range van der Waals Forces between Protons *Physics Letters B* [P-BARUT-W0175]

200. **1975** Radiation Damping in a Gravitational Field *Journal of Physics A* [P-BARUT-W0176]

201. **1975** Radiation Reaction and Mass Renormalization in Scalar and Tensor Fields and Linearized Gravitation *Journal of Physics A* [P-BARUT-W0177]

202. **1975** Resonances in e+e- System Due to Anomalous Magnetic Moment Interactions *Physics Letters B* [P-BARUT-W0178]

203. **1975** The Physics of Magnetic Monopoles *Physikalische Blätter* [P-BARUT-W0179]

204. **1976** 50.Yılında Schrödinger Denklemi; Güncel Araştırma Problemlerine Bakış *Çağdaş Fizik, c.1, sayı 2 (Kasım 1976), s.9–15* [P-BARUT-W0195]

205. **1976** Angular Momentum of Systems of Electric and Magnetic Charges and of Singular Flux Surfaces *Journal of Mathematical Physics* [P-BARUT-W0180]

206. **1976** Electrodynamics with Strings and Monopoles [P-BARUT-W0181]

207. **1976** Inelastic Transition Form Factors in H-Atom *Physical Review A* [P-BARUT-W0561]

208. **1976** Infinite Component Fields. I. Electromagnetic Inelastic Form Factors and Structure Functions of the Proton *Physical Review D* [P-BARUT-W0182]

209. **1976** Infinite Component Fields. II. The Electromagnetic Structure of Bosons *Physical Review D* [P-BARUT-W0183]

210. **1976** Magnetic Charge and Hyperfine Splitting in Hydrogen *Physics Letters B* [P-BARUT-W0184]

211. **1976** Mass Spectrum in Relativistic Extended Models of Elementary Particles *Letters in Mathematical Physics* [P-BARUT-W0185]

212. **1976** New Theory of Positronium Decay and Relativistic Wave Function Corrections *Physics Letters A* [P-BARUT-W0186]

213. **1976** Nonlinear O(n+1)-Symmetric Field Theories, Symmetry Breaking and Finite Energy Solutions *Helvetica Physica Acta* [P-BARUT-W0187]

214. **1976** Quantized Excitations of Relativistic Extended Elementary Particles *Il Nuovo Cimento B* [P-BARUT-W0188]

215. **1976** Solution of the Dirac Equation with Coulomb and Magnetic Moment Interactions *Journal of Mathematical Physics* [P-BARUT-W0189]

1926 GENERATION EVIDENCE ATLAS
RECONSTRUCTED - PROJECT DOCUMENT
NOT AN ORIGINAL HISTORICAL DOCUMENT

216. **1976** Some New Identities of Clebsch-Gordan Coefficients and Representation Functions of SO(2,1) and SO(4) *Journal of Mathematical Physics* [P-BARUT-W0190]

217. **1976** Space-Like States in Relativistic Quantum Theory [P-BARUT-W0191]

218. **1976** Structure of Space-Time in the Small: Electromagnetic Fields over Manifolds with Betti Numbers [P-BARUT-W0192]

219. **1976** The Conformally Invariant Action-at-a-Distance Electrodynamics *Journal of Physics A* [P-BARUT-W0193]

220. **1976** The Electromagnetic Polarizabilities of Nucleons *Physical Review D* [P-BARUT-W0194]

221. **1977** Beyond Symmetry, Homeometry and Homeometry Groups *Topics in Mathematical Physics* [P-BARUT-W0564]

222. **1977** Characteristic 'Quanta' of Nonlinear Field Equations *Nonlinear Equations in Physics and Mathematics (Istanbul 1977 proceedings)* [P-BARUT-W0196]

223. **1977** Charge Quantization Condition with n Strings. A New Internal Quantum Number of Charge-Monopole Systems *Letters in Mathematical Physics* [P-BARUT-W0197]

224. **1977** De Rham Currents, Extended Singularities of Fields and Magnetic Monopoles *Reports on Mathematical Physics* [P-BARUT-W0198]

225. **1977** Dynamical Group Quantization *Reports on Mathematical Physics* [P-BARUT-W0199]

226. **1977** Electromagnetic Fields on Manifolds: Betti Numbers, Monopoles and Strings, Minimal Coupling *Differential Geometrical Methods in Mathematical Physics II (Bonn 1977 proceedings)* [P-BARUT-W0200]

227. **1977** Elementary Particles, Universes and Singularity Surfaces *Conference on Cosmology, History and Theology* [P-BARUT-W0562]

228. **1977** Form-Factor Corrections to Superpositronium and Short Distance Behavior of Magnetic Moment of the Electron *Physical Review D* [P-BARUT-W0201]

229. **1977** Interpretation of Space-Like Solutions of Infinite Component Wave Equations and Grodsky-Streater No-Go Theorem *Journal of Physics A* [P-BARUT-W0202]

230. **1977** New Dynamical Phenomena in Nonperturbative Quantum Electrodynamics [P-BARUT-W0203]

231. **1977** Perturbation Theory for Infinite Component Wave Equations *Journal of Physics A* [P-BARUT-W0204]

232. **1977** Schrödinger Equation—Fifty Years Later. A Survey of Current Research Problems *Zeitschrift für Naturforschung* [P-BARUT-W0563]

233. **1977** Soliton Solutions of Infinite Component Wave Equations *Physics Letters B* [P-BARUT-W0205]

234. **1977** Theory of Group Representations and Applications *Polish Scientific Publishers* [P-BARUT-W0206]

235. **1978** Consequences of a Dynamical Theory of the Planck's Constant h/2π for the Origin of Quantum Theory *Quantum Theory and the Structure of Time and Space, Vol. 3 (Tutzing 1978 proceedings)* [P-BARUT-W0207]

236. **1978** Covariant Field Equations for an Electromagnetic String with Mass Point at the Ends *Journal of Physics A* [P-BARUT-W0208]

237. **1978** Meson Spectroscopy Based on Proper Oscillations of Nonlinear Fields: An Equivalence Theorem *Hadronic Journal* [P-BARUT-W0209]

238. **1978** Nonlinear Equations in Physics and Mathematics. Proceedings, NATO Advanced Study Institute, Istanbul, Turkey, August 1-13, 1977 [P-BARUT-W0210]

239. **1978** Nonlinear Problems in Classical and Quantum Electrodynamics *Nonlinear Problems in Theoretical Physics (Jaca 1978 proceedings)* [P-BARUT-W0211]

240. **1978** Solutions of Nonlinear Equations Simulating Pair Production and Pair Annihilation *Physics Letters A* [P-BARUT-W0212]

241. **1978** The Creation of a Photon: Calculation of the Planck's Constant ħ or the Fine Structure Constant α [P-BARUT-W0213]

242. **1978** The Erratic Electron *Topics in Nonlinear Dynamics, AIP Conference Proceedings 46* [P-BARUT-W0565]

243. **1978** The Mass of the Muon *Physics Letters B* [P-BARUT-W0214]

244. **1979** Bäcklund Transformations and New Solutions of Nonlinear Wave Equations in Four-Dimensional Space-Time *Letters in Mathematical Physics* [P-BARUT-W0215]

245. **1979** Charge Quantization Theorem. Reply to a Paper by Usachev *Soviet Journal of Particles and Nuclei* [P-BARUT-W0216]

246. **1979** Gauge Invariant Formulation of Dyonium Hamiltonian on the Sphere S^3 *Lettere al Nuovo Cimento* [P-BARUT-W0217]

247. **1979** Gravitation and Electromagnetism *General Relativity, Part A (Trieste 1979 proceedings)* [P-BARUT-W0218]

248. **1979** Group-Theoretical Equivalence of Quantum Mechanical Systems *Lie-Bäcklund Transformations and Applications* [P-BARUT-W0567]

249. **1979** Hadronic Multiplets in Terms of Absolutely Stable Particles: An Already Unified Theory *Group Theoretical Methods in Physics (Kiryat Anavim 1979 proceedings)* [P-BARUT-W0219]

250. **1979** How to Build Hadron Multiplets from Stable Particles *Lecture Notes in Physics* [P-BARUT-W0220]

251. **1979** Hydrogenium and Hydrogenium Ion [P-BARUT-W0221]

252. **1979** Infinite Component Wave Equations Describe Relativistic Composite Systems *Group Theoretical Methods in Physics (Kiryat Anavim 1979 proceedings)* [P-BARUT-W0222]

253. **1979** Kℓ3 Form Factors *Physical Review D* [P-BARUT-W0223]

254. **1979** Lepton Mass Formula *Physical Review Letters* [P-BARUT-W0224]

255. **1979** On Electrodynamics with Internal Fermionic Excitations *Lettere al Nuovo Cimento* [P-BARUT-W0225]

256. **1979** Quantum Theory of Infinite Component Fields *Journal of Mathematical Physics* [P-BARUT-W0226]

257. **1979** Renormalizable Field Theories with Multiple Fermionic Excitations *Physics Letters B* [P-BARUT-W0227]

258. **1979** Some New Physical Inputs in Quantum Electrodynamics *XVth Karpacz Winter School on Mathematical Physics* [P-BARUT-W0566]

259. **1979** The Borderline between Classical Physics and Quantum Theory: Towards a Theory of the Planck's Constant *Czechoslovak Journal of Physics B* [P-BARUT-W0228]

260. **1979** The Theorem of Charge Quantization *Fizika Elementarnykh Chastits i Atomnogo Yadra* [P-BARUT-W0229]

1928 GENERATION EVIDENCE ATLAS
RECONSTRUCTED PROJECT DOCUMENT
NOT AN ORIGINAL HISTORICAL DOCUMENT

261. **1979** Two-Fermion Equation with Form Factors and Electromagnetic High Mass Resonances *Acta Physica Polonica B* [P-BARUT-W0230]

262. **1979** What Are the True Building Blocks of Matter *Symmetries in Science (Carbondale 1979 proceedings)* [P-BARUT-W0231]

263. **1980** A Dynamical Model for the Two-Component Pomeron *Lettere al Nuovo Cimento* [P-BARUT-W0232]

264. **1980** Completely Integrable N-Body Problems in Three Dimensions and Their Relativistic Generalization *Group Theoretical Methods in Physics (Cocoyoc 1980 proceedings)* [P-BARUT-W0233]

265. **1980** Dynamical Groups for the Motion of Relativistic Composite Systems *Groups, Systems and Many-Body Physics* [P-BARUT-W0234]

266. **1980** Electrodynamics and Classical Theory of Fields and Particles *Dover* [P-BARUT-W0235]

267. **1980** Electromagnetic Interactions Beyond Quantum Electrodynamics *Foundations of Radiation Theory and Quantum Electrodynamics* [P-BARUT-W0236]

268. **1980** Foundations of Radiation Theory and Quantum Electrodynamics [P-BARUT-W0237]

269. **1980** Lectures on Magnetic Interactions of Stable Particles and Magnetic Resonances *Group Theory and Its Applications in Physics-1980 (Mexico City 1980 proceedings)* [P-BARUT-W0238]

270. **1980** Magnetic Resonances between Massive and Massless Spin 1/2 Particles with Magnetic Moments *Journal of Mathematical Physics* [P-BARUT-W0239]

271. **1980** Narrow Resonances as an Eigenvalue Problem and Applications to High-Energy Magnetic Resonances: An Exactly Soluble Model *Journal of Mathematical Physics* [P-BARUT-W0240]

272. **1980** Return to 1932: Proton, Electron and Neutrino as True Elementary Constituents of Leptons, Hadrons and Nuclei *Quantum Theory and the Structure of Time and Space, Vol. 4 (Tutzing 1980 proceedings)* [P-BARUT-W0241]

273. **1980** Stable Particles as Building Blocks of Matter *Surveys in High Energy Physics* [P-BARUT-W0242]

274. **1980** The Flavor Sequence and Superselection Rules *Group Theory and Its Applications in Physics-1980 / Mexico Group Theory 1980 proceedings (index label 'Mexico Grp.Theor.1980')* [P-BARUT-W0243]

275. **1980** The Magnetic Moment of the Neutrino *eConf C801002* [P-BARUT-W0244]

276. **1980** The Zitterbewegung of the Electron and Its Compact Phase Space *Group Theoretical Methods in Physics (Cocoyoc 1980 proceedings)* [P-BARUT-W0245]

277. **1980** Theory of Group Representations and Applications *Polish Scientific Publishers* [P-BARUT-W0246]

278. **1981** Completely Integrable N-Body Quantum Systems in 3-Dimension I *Journal of Physics A* [P-BARUT-W0570]

279. **1981** Conformal Covariance and the Probability Interpretation of Wave Equations *Physics Letters A* [P-BARUT-W0247]

280. **1981** Derivation and Uniqueness of Vacuum Solutions of Conformally Covariant Coupled Nonlinear Field Equations *Physics Letters B* [P-BARUT-W0248]

281. **1981** Magnetic Moment Operator of the Relativistic Electron *Physical Review D* [P-BARUT-W0249]

282. **1981** New Type of Vacuum Solutions of Nonlinear Conformally Covariant Coupled Field Equations *Physical Review D* [P-BARUT-W0250]

283. **1981** On the Dynamical Group of the Charge-Monopole System *Journal of Physics A* [P-BARUT-W0251]

284. **1981** The Phenomenon of Magnetic Pairing and Exactly Soluble Models of Magnetic Resonances *Quantum Mechanics in Mathematics, Chemistry and Physics* [P-BARUT-W0569]

285. **1981** The Wave Equation for the Electron-Muon System and Their Neutrinos *Physics Letters B* [P-BARUT-W0252]

286. **1981** The Zitterbewegung and the Internal Geometry of the Electron *Physical Review D* [P-BARUT-W0253]

287. **1982** Bemerkungen über C. F. von Weizsäckers Theorie der Kernkräfte *Physik, Philosophie und Politik* [P-BARUT-W0574]

288. **1982** Completely Integrable N-Body Quantum Systems in 3-Dimension II *Journal of Physics A* [P-BARUT-W0572]

289. **1982** Composite Models of Pion and Muon and a Calculation of the Fermi Coupling Constant from Their Decay *Kinam* [P-BARUT-W0254]

290. **1982** Connection between the Stable Particle Model and the Integrally Charged Quark Model *Lettere al Nuovo Cimento* [P-BARUT-W0255]

291. **1982** Derivation of Infinite Component Wave Equations from Field Theory *Group Theoretical Methods in Physics (Istanbul 1982 proceedings)* [P-BARUT-W0256]

292. **1982** Derivation of Static Pauli Spin-Orbit and Spin-Spin Potentials from Field Theory *Physica Scripta* [P-BARUT-W0257]

293. **1982** Derivation of Strong and Weak Forces from Magnetic Interactions in Quantum Electrodynamics (QED) *Czechoslovak Journal of Physics B* [P-BARUT-W0258]

294. **1982** Description and Interpretation of the Internal Symmetries of Hadrons as an Exchange Symmetry *Physica A* [P-BARUT-W0259]

295. **1982** Electrodynamics in Terms of Functions over the Group SU(2). II. Quantization *Foundations of Physics* [P-BARUT-W0260]

296. **1982** Exact Solutions of the Heisenberg Equations and Zitterbewegung of the Electron in a Constant Uniform Magnetic Field *Australian Journal of Physics* [P-BARUT-W0267]

297. **1982** Meron-Like Solutions of Conformally Covariant Coupled Nonlinear Field Equations *Physica D* [P-BARUT-W0261]

298. **1982** On Conformally Covariant Spin-2 and Spin-3/2 Equations *Journal of Physics A* [P-BARUT-W0262]

299. **1982** On the Shape of Tachyons *Il Nuovo Cimento A* [P-BARUT-W0263]

300. **1982** Particle Physics Based on Stable Particles as Constituents *Quantum Theory and the Structure of Time and Space, Vol. 5 (Tutzing 1982 proceedings)* [P-BARUT-W0264]

301. **1982** Relativistic Formula for the Magnetic Part of the Lamb Shift and Its Z-Dependence *Physica Scripta* [P-BARUT-W0573]

302. **1982** Schrödinger Equations with SU(1,1)-Dynamical Symmetry *Journal of Boğaziçi University* [P-BARUT-W0576]

303. **1982** Temel Bilimler: Dün, Bugün ve Yarın *Çağdaş Fizik, sayı 14 (Kasım 1982), s.3–4* [P-BARUT-W0269]

304. **1982** Upper Limit on the Magnetic Moment of the Neutrino *Physical Review D* [P-BARUT-W0265]

1996 GENERATION EVIDENCE ATLAS
RECONSTRUCTED - PROJECT DOCUMENT
NOT AN ORIGINAL LHCT CRUCIAL DOCUMENT

305. **1982** What Kind of a Dynamical System Is the Radiating Electron? *Lecture Notes in Mathematics* [P-BARUT-W0266]

306. **1983** A Duality between Massless Particles and a Charge-Monopole System *Letters in Mathematical Physics* [P-BARUT-W0270]

307. **1983** A Family of Groups Generalizing the Poincaré Group and Their Physical Applications *Journal of Physics A: Mathematical and General* [P-BARUT-W0271]

308. **1983** A New Representation of Magnetic Sources in Field Theory and Its Relation to Pauli and Fermi Couplings *Physics Letters B* [P-BARUT-W0272]

309. **1983** Capture of a Dirac Monopole by a Magnetic Dipole *Physical Review D* [P-BARUT-W0273]

310. **1983** Derivation of Relativistic Charge-Monopole and Monopole-Monopole Potentials from Field Theory *Annals of Physics* [P-BARUT-W0274]

311. **1983** Elementary Particle States Based on the Clifford Algebra C7 *International Journal of Theoretical Physics* [P-BARUT-W0275]

312. **1983** Group Action on Homogeneous Spaces and Solutions of Nonlinear Dynamical System *Conference on Differential Geometric Methods in Theoretical Physics* [P-BARUT-W0577]

313. **1983** Large Magnetic Moment Effects at High Energies and Composite Particle Models *Field Theory in Elementary Particles* [P-BARUT-W0277]

314. **1983** Negative-Norm States, Superselection Rules, and the Lepton Family *Physical Review D* [P-BARUT-W0278]

315. **1983** Neutrino and Antineutrino Elastic Scattering and Photoneutrino Reactions *Physical Review D* [P-BARUT-W0279]

316. **1983** Nonperturbative Quantum Electrodynamics: The Lamb Shift *Foundations of Physics* [P-BARUT-W0280]

317. **1983** Nonperturbative Treatment of Magnetic Interactions at Short Distances and a Simple Magnetic Model of Matter *Quantum Electrodynamics of Strong Fields* [P-BARUT-W0281]

318. **1983** On Conservation of Parity and Time Reversal and Composite Models of Particles *Foundations of Physics* [P-BARUT-W0282]

319. **1983** Period Doubling in the (n+l) Filling Rule and Dynamical Symmetry of the Demkov-Ostrovsky Atomic Model *Journal of Physics B: Atomic and Molecular Physics* [P-BARUT-W0284]

320. **1983** Quantum Theory, Groups, Fields and Particles *Mathematical Physics Studies* [P-BARUT-W0285]

321. **1983** The Relativistic Oscillator: Linearly Rising Trajectories and Structure Functions *Physical Review Letters* [P-BARUT-W0286]

322. **1983** The Zitterbewegung and the Einstein A Coefficient of Spontaneous Emission *Old and New Questions in Physics, Cosmology, Philosophy, and Theoretical Biology* [P-BARUT-W0287]

323. **1983** Theory of Dynamical Systems Bound by Magnetic Forces *Quantum Theory, Groups, Fields and Particles* [P-BARUT-W0288]

324. **1983** Transformation between Fractionally Charged and Integrally Charged Quarks Induced by Rishons *Lettere al Nuovo Cimento* [P-BARUT-W0290]

325. **1983** What Is an Electron? Relativistic Electron Theory and Radiative Processes *Quantum Optics, Experimental Gravity, and Measurement Theory* [P-BARUT-W0291]

326. **1984** A Classical Model of EPR Experiment with Quantum Mechanical Correlations and Bell Inequalities *Physics Letters A* [P-BARUT-W0292]

327. **1984** A Generalized Local U(1) Gauge Invariance for Both the Standard and Pauli Couplings of QED *Physics Letters B* [P-BARUT-W0293]

328. **1984** Classical Model of the Dirac Electron *Physical Review Letters* [P-BARUT-W0294]

329. **1984** Complex-Energy Solutions of the Relativistic Kepler Problem at High Energies *Physical Review D* [P-BARUT-W0295]

330. **1984** Nonperturbative Quantum Electrodynamics of Bound States and Radiative Processes *Quantum Electrodynamics and Quantum Optics* [P-BARUT-W0296]

331. **1984** On Bell's Inequalities *Acta Physica Austriaca* [P-BARUT-W0579]

332. **1984** On the Dynamical Symmetry of the Periodic Table. II. Modified Demkov-Ostrovsky Atomic Model *Journal of Physics B: Atomic and Molecular Physics* [P-BARUT-W0297]

333. **1984** Path-Integral Derivation of the Dirac Propagator *Physical Review Letters* [P-BARUT-W0298]

334. **1984** Quantum Electrodynamics and Quantum Optics. Proceedings, NATO Advanced Study Institute, Boulder, USA, May 27 - June 8, 1983 *NATO ASI Series B* [P-BARUT-W0299]

335. **1984** Quantum, Space and Time: The Quest Continues [P-BARUT-W0300]

336. **1984** Relativistic Electron Theory and Foundations of Quantum Theory *International Symposium Foundations of Quantum Mechanics, Tokyo 1983* [P-BARUT-W0578]

337. **1984** Relativistic Nonperturbative Treatment of Spin Forces *AIP Conference Proceedings* [P-BARUT-W0301]

338. **1984** Rotational Invariance, Locality, and Einstein-Podolsky-Rosen Experiments *Physical Review Letters* [P-BARUT-W0302]

339. **1984** The Zitterbewegung of the Neutrino *Letters in Mathematical Physics* [P-BARUT-W0303]

340. **1984** Vacuum Solutions of Conformally Covariant Nonlinear Coupled Equations (in Chinese) *HEPNP* [P-BARUT-W0304]

341. **1985** Bound States, Resonances, and Symmetries of a Neutral Dirac Particle with Anomalous Magnetic Moment, Coupled to a Fixed Monopole *Journal of Mathematical Physics* [P-BARUT-W0305]

342. **1985** Classical and Quantum Spin Correlations: Generalization to Triplet and Other States *Symposium on the Foundations of Modern Physics* [P-BARUT-W0583]

343. **1985** Compact Quantum Systems: Internal Geometry of Relativistic Systems *Journal of Mathematical Physics* [P-BARUT-W0306]

344. **1985** Covariant Generalization of the Zitterbewegung of the Electron and Its SO(4,2) and SO(3,2) Internal Algebras *Physical Review D* [P-BARUT-W0307]

345. **1985** Derivation of Nonperturbative Relativistic Two-Body Equations from the Action Principle in Quantum Electrodynamics *Fortschritte der Physik* [P-BARUT-W0308]

346. **1985** Exact Solutions of the Heisenberg Equations of the Relativistic Electron in an External Electromagnetic Wave *Acta Physica Austriaca* [P-BARUT-W0309]

347. **1985** Magnetic Selffocusing Principle and Electromagnetic Model of Matter *Hadronic Journal Supplement* [P-BARUT-W0310]

1926 GENERATION EVIDENCE ATLAS
RECONSTRUCTED - PROJECT DOCUMENT
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348. **1985** Mass Formula for Z0 and Other Predictions *Lettere al Nuovo Cimento* [P-BARUT-W0311]

349. **1985** Massive Narrow Resonance Solutions of Strongly Coupled Relativistic Particles: A Mechanism of Mass Generation *Letters in Mathematical Physics* [P-BARUT-W0312]

350. **1985** On the Connection between Dyons, Vortices, and Aharonov-Bohm Effect *Annals of Physics* [P-BARUT-W0313]

351. **1985** On the Dynamical Group of the Kepler Problem in a Curved Space of Constant Curvature *Physics Letters A* [P-BARUT-W0582]

352. **1985** On the Gyromagnetic Ratio in the Kaluza-Klein Theories and the Schuster-Blackett Law *Foundations of Physics* [P-BARUT-W0314]

353. **1985** Photoproduction of a Neutrino Pair from an Electron and Astrophysical Implications of Neutrino Emission Processes *Physical Review D* [P-BARUT-W0315]

354. **1985** Quantum Electrodynamics Based on Selfenergy: Lamb Shift and Spontaneous Emission without Field Quantization *Physical Review A* [P-BARUT-W0316]

355. **1985** Radial Equations for the Relativistic Two Fermion Problem with the Most General Electric and Magnetic Potentials *Fortschritte der Physik* [P-BARUT-W0317]

356. **1985** Relativistic Quantum Theory of Fermions Based on the Clifford Algebra C(7) *Journal of Mathematical Physics* [P-BARUT-W0318]

357. **1985** Solutions of the Basic Problems of Electrodynamics in the Group Space-Formulation *Il Nuovo Cimento B* [P-BARUT-W0580]

358. **1985** The Zitterbewegung of the Electron in External Fields *Physical Review D* [P-BARUT-W0581]

359. **1986** An Exactly Soluble Relativistic Quantum Two Fermion Problem *Journal of Mathematical Physics* [P-BARUT-W0319]

360. **1986** Bilim Adamı Yetiştirmek İçin Seferberlik *Çağdaş Fizik, sayı 18 (Kasım 1986), s.6–7* [P-BARUT-W0327]

361. **1986** Center-of-Mass Motion of a System of Relativistic Dirac Particles *Few-Body Systems* [P-BARUT-W0320]

362. **1986** Conformal Groups and Related Symmetries: Physical Results and Mathematical Background. Proceedings of a Symposium Held at the Arnold Sommerfeld Institute for Mathematical Physics, Technical University of Clausthal, Germany, August 12-14, 1985 *Lecture Notes in Physics* [P-BARUT-W0321]

363. **1986** From Heisenberg Algebra to Conformal Dynamical Group *Conformal Groups and Related Symmetries: Physical Results and Mathematical Background (Lecture Notes in Physics 261)* [P-BARUT-W0322]

364. **1986** Nonperturbative Quantum Electrodynamics without Infinities *ICTP preprint* [P-BARUT-W0326]

365. **1986** Quantum Electrodynamics Based on Self-Energy Versus Quantization of Fields: Illustration by a Simple Model *Physical Review A* [P-BARUT-W0586]

366. **1986** The Four States of the Massless Neutrino with Pauli Coupling by Spin Gauge Invariance *Letters in Mathematical Physics* [P-BARUT-W0323]

367. **1986** The Lagrange Triangle and the Model of the Proton *Journal of the Technical University of Istanbul, Special Issue* [P-BARUT-W0587]

368. **1986** Theoretical Experiments on the Foundations of Quantum Theory *Fundamental Questions in Quantum Mechanics* [P-BARUT-W0584]

369. **1986** Theory of Group Representations and Applications *World Scientific Publishing Company* [P-BARUT-W0324]

370. **1986** Three Lectures on the Foundations of Quantum Theory and Quantum Electrodynamics *Frontiers in Nonequilibrium Statistical Mechanics* [P-BARUT-W0585]

371. **1986** Unification Based on Electromagnetism: A Simple Composite Model of Particles *Annalen der Physik* [P-BARUT-W0325]

372. **1987** A New Approach to Bound-State Quantum Electrodynamics. I. Theory *Physica A: Statistical Mechanics and its Applications* [P-BARUT-W0339]

373. **1987** A New Approach to Bound-State Quantum Electrodynamics. II. Spectra of Positronium, Muonium and Hydrogen *Physica A: Statistical Mechanics and its Applications* [P-BARUT-W0340]

374. **1987** A New Realization of Dynamical Groups and Factorization Method *Journal of Physics A: Mathematical and General* [P-BARUT-W0346]

375. **1987** A Universal Law for the Increase of Hadronic Cross-Sections at High Energies *Hadronic Journal* [P-BARUT-W0328]

376. **1987** Algebraic Treatment of Second Pöschl-Teller, Morse-Rosen and Eckart Equations *Journal of Physics A: Mathematical and General* [P-BARUT-W0347]

377. **1987** An Approach to Finite Non-Perturbative Quantum Electrodynamics *2nd International Symposium on Foundations of Quantum Mechanics* [P-BARUT-W0589]

378. **1987** Classical Model of the Dirac Electron in Curved Space *Classical and Quantum Gravity* [P-BARUT-W0341]

379. **1987** Electron as a Radiating and Spinning Dynamical System and Discrete Internal Quantum Systems *Physica Scripta* [P-BARUT-W0336]

380. **1987** Exact Solutions of the Dirac Equation in Spatially Flat Robertson-Walker Space-Times *Physical Review D* [P-BARUT-W0350]

381. **1987** Irreversibility, Organization, and Self-Organization in Quantum Electrodynamics *Foundations of Physics* [P-BARUT-W0342]

382. **1987** Large Polarization Asymmetries in Nucleon-Nucleon Scattering *Fundamental Symmetries* [P-BARUT-W0329]

383. **1987** Magnetic Resonances and the Positron Peak in Heavy-Ion Collisions *Physics of Strong Fields* [P-BARUT-W0330]

384. **1987** Nonperturbative Radiative and Bound-State Equations for Strong Quantum Electrodynamics *Physics of Strong Fields* [P-BARUT-W0331]

385. **1987** On the Connection between CP-Violation, Muon-Neutrino Lifetime, Muonium Conversion and K0-Decays: An Explicit Model *Fundamental Symmetries* [P-BARUT-W0332]

386. **1987** On the Strong van der Waals Forces between Hadrons Due to Confining Potentials *International Journal of Modern Physics A* [P-BARUT-W0335]

387. **1987** On the Theory of the Stern-Gerlach Apparatus *Foundations of Physics* [P-BARUT-W0343]

388. **1987** On the Treatment of Møller and Breit Potentials and the Covariant Two-Body Equation for Positronium and Muonium *Physica Scripta* [P-BARUT-W0348]

389. **1987** Path Integral Treatment of the Hydrogen Atom in a Curved Space of Constant Curvature *Journal of Physics A: Mathematical and General* [P-BARUT-W0351]

390. **1987** Quantization of the Classical Relativistic Zitterbewegung in the Schrödinger Picture *Classical and Quantum Gravity* [P-BARUT-W0344]

1920 GENERATION EVIDENCE ATLAS
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391. **1987** Quantum Electrodynamics Based on Self-Energy, without Second Quantization: The Lamb Shift and Long-Range Casimir-Polder van der Waals Forces near Boundaries *Physical Review A* [P-BARUT-W0349]

392. **1987** Quantum Electrodynamics Based on Self-Energy: Spontaneous Emission in Cavities *Physical Review A* [P-BARUT-W0345]

393. **1987** The Generalized Morse Oscillator in the SO(4,2) Dynamical Group Scheme *Journal of Mathematical Physics* [P-BARUT-W0337]

394. **1987** The Remarkable Phase Space of the Radiating Electron *The Physics of Phase Space: Nonlinear Dynamics and Chaos, Geometric Quantization, and Wigner Function (Lecture Notes in Physics)* [P-BARUT-W0333]

395. **1987** Towards a Theory of Single Events in Spin Correlation Experiments *New Techniques and Ideas in Quantum Measurement Theory* [P-BARUT-W0588]

396. **1987** Tunnelling of Neutral Spin-1/2 Particles through Magnetic Fields *Zeitschrift für Physik A Atomic Nuclei* [P-BARUT-W0338]

397. **1988** Can We Calculate the Fundamental Dimensionless Constants of Physics? *Gravitational Measurements, Fundamental Metrology and Constants* [P-BARUT-W0352]

398. **1988** Classical and Quantum 'EPR' Spin Correlations in the Triplet State *Il Nuovo Cimento B* [P-BARUT-W0370]

399. **1988** Combining Relativity and Quantum Mechanics: Schrödinger's Interpretation of ψ *Foundations of Physics* [P-BARUT-W0363]

400. **1988** Comment on the Paper 'An Exactly Soluble Relativistic Quantum Two-Fermion Problem' *Journal of Mathematical Physics* [P-BARUT-W0364]

401. **1988** Conformally Covariant Coupled Non-Linear Field Theory on the Hypercone: Vacuum Solutions and Quantization of Normal Modes *Modern Physics Letters A* [P-BARUT-W0365]

402. **1988** Covariant Generalization of the Zitterbewegung of the Electron and Its SO(4,2) and SO(3,2) Internal Algebras *Dynamical Groups and Spectrum Generating Algebras* [P-BARUT-W0353]

403. **1988** Dynamical Groups and Mass Formula *Dynamical Groups and Spectrum Generating Algebras* [P-BARUT-W0354]

404. **1988** Dynamical Groups and Spectrum Generating Algebras [P-BARUT-W0355]

405. **1988** Dynamical Groups for the Motion of Relativistic Composite Systems *Dynamical Groups and Spectrum Generating Algebras* [P-BARUT-W0356]

406. **1988** Erwin Schrödinger (1887-1961)'in 100. yıldönümü ve kuantum mekaniğinin bugünkü durumu *Çağdaş Fizik Dergisi, sayı 20, Kasım 1988, s. 13-16* [P-BARUT-W0376]

407. **1988** Explicit Calculations with a Hidden-Variable Spin Model *Quantum Mechanics Versus Local Realism* [P-BARUT-W0357]

408. **1988** Joint Probabilities of Noncommuting Operators and Incompleteness of Quantum Mechanics *Foundations of Physics* [P-BARUT-W0375]

409. **1988** Kaluza-Klein Approach to the Classical Model of the Dirac Electron *Classical and Quantum Gravity* [P-BARUT-W0371]

410. **1988** Lorentz-Dirac Equation and Energy Conservation for Radiating Electrons *Physics Letters A* [P-BARUT-W0373]

411. **1988** On de Broglie's "Méthode de Fusion" for Higher Spin Particles *Louis de Broglie, Que Nous Avons Connu* [P-BARUT-W0591]

412. **1988** Passage of Relativistic Magnetic Dipole Moments through Magnetic Fields *Zeitschrift für Physik A Atomic Nuclei* [P-BARUT-W0374]

413. **1988** Path Integration via Hamilton-Jacobi Coordinates and Applications to Potential Barriers *Physical Review A* [P-BARUT-W0379]

414. **1988** Quantum Electrodynamics Based on Self-Fields, without Second Quantization: A Nonrelativistic Calculation of g-2 *Physical Review A* [P-BARUT-W0377]

415. **1988** Quantum-Electrodynamics Based on Self-Energy *Physica Scripta* [P-BARUT-W0366]

416. **1988** Reduction of a Class of O(4,2) Representations with Respect to SO(4,1) and SO(3,2) *Dynamical Groups and Spectrum Generating Algebras* [P-BARUT-W0358]

417. **1988** Relativistic Theory of Spontaneous Emission *Physical Review A* [P-BARUT-W0369]

418. **1988** SO(4,2)-Formulation of the Symmetry Breaking in Relativistic Kepler Problems with or without Magnetic Charges *Dynamical Groups and Spectrum Generating Algebras* [P-BARUT-W0360]

419. **1988** Schrödinger's Interpretation of Ψ as a Continuous Charge Distribution *Annalen der Physik* [P-BARUT-W0367]

420. **1988** Selected Scientific Papers of Alfred Landé *Fundamental Theories of Physics* [P-BARUT-W0359]

421. **1988** Some Unusual Applications of Lie Algebra Representations in Quantum Theory *Dynamical Groups and Spectrum Generating Algebras* [P-BARUT-W0361]

422. **1988** The Revival of Schrödinger's Interpretation of Quantum Mechanics *Foundations of Physics Letters* [P-BARUT-W0368]

423. **1988** The Spinning Minimal Surfaces without the Grassmann Variables *Letters in Mathematical Physics* [P-BARUT-W0378]

424. **1988** The Tacit Postulate on Single Events and the Comparison of Classical and Quantum Spin Correlation Experiments *Microphysical Reality and Quantum Formalism* [P-BARUT-W0362]

425. **1988** Two- and Three-Dimensional Tunnelling of Neutral Spin Particles through Magnetic Fields *Physica B+C* [P-BARUT-W0372]

426. **1989** Algebraic Scattering Theory of Relativistic Composite Particles *Fortschritte der Physik* [P-BARUT-W0380]

427. **1989** Analytic Group-Theoretical Form Factors of Hydrogenlike Atoms for Discrete and Continuum Transitions *Physical Review A* [P-BARUT-W0381]

428. **1989** Angular Distribution in Electron-Neutrino Scattering and the Anomalous Magnetic Moment of the Neutrino *Il Nuovo Cimento A* [P-BARUT-W0382]

429. **1989** Derivation of the Envelope (de Broglie) Solitons for Nonlinear Klein-Gordon Equations by Hirota's Method *Journal of the Physical Society of Japan* [P-BARUT-W0595]

430. **1989** Equivalence of the Spinning Superparticle Descriptions with Grassmann Variables or with C Number Spinors *Physics Letters B* [P-BARUT-W0383]

431. **1989** Foundations of self-field quantum electrodynamics *Istanbul 1989, New Frontiers in Quantum Electrodynamics and Quantum Optics* [P-BARUT-W0384]

432. **1989** Generalization of the Lorentz-Dirac Equation to Include Spin *Physical Review A* [P-BARUT-W0385]

433. **1989** Geometry and Physics: Non-Newtonian Forms of Dynamics *Monographs and Textbooks in Physical Science / Lecture Notes* [P-BARUT-W0386]

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434. **1989** Indecomposable Representations of Lie Algebras and Lie Groups *Aynı eser* [P-BARUT-W0599]

435. **1989** Membranes, strings, extended elementary particles as singularities of fields *Trieste 1989, Supermembranes and Physics in 2+1 Dimensions* [P-BARUT-W0387]

436. **1989** On the Wave Particle-Like Solutions of Nonlinear Equations *Canadian Journal of Physics* [P-BARUT-W0388]

437. **1989** Path Integral Formulation of Quantum Electrodynamics from Classical Particle Trajectories *Physics Reports* [P-BARUT-W0389]

438. **1989** Physical Application of Indecomposable Representations *Indecomposable Representations* [P-BARUT-W0598]

439. **1989** Problème Relativiste à deux Corps en Electrodynamique Quantique *Helvetica Physica Acta* [P-BARUT-W0596]

440. **1989** QED Based on Selffields: A Relativistic Calculation of g-2 *Zeitschrift für Naturforschung A* [P-BARUT-W0390]

441. **1989** QED in nonsimply connected regions [P-BARUT-W0391]

442. **1989** QED: The unfinished business *Islamabad 1989, Mathematical Physics* [P-BARUT-W0392]

443. **1989** Quantum Electrodynamics Based on Selffields, without Second Quantization: Apparatus Dependent Contributions to g-2 *Physical Review A* [P-BARUT-W0393]

444. **1989** The Electromagnetic Field as Oscillations of a Relativistic Deformable Ether *Recent Advances in Engineering Science: C. Eringen Symposium* [P-BARUT-W0593]

445. **1989** The Schrödinger and the Dirac Equation: Linear, Nonlinear and Integrodifferential [P-BARUT-W0394]

446. **1989** Translational and Internal Motions of Classical and Quantum Spins in Passing through Magnetic Fields *Spin Polarized Quantum Spin Systems* [P-BARUT-W0594]

447. **1990** A Model for Polarization Asymmetries in pp-Scattering *Spin and Polarization Dynamics in Nuclear and Particle Physics* [P-BARUT-W0601]

448. **1990** E= ω *Physics Letters A* [P-BARUT-W0403]

449. **1990** Excited states of zitterbewegung *Physics Letters B* [P-BARUT-W0405]

450. **1990** Interpretation of Quantum Potential for Stationary States and Causal Trajectories *Annales de la Fondation Louis de Broglie* [P-BARUT-W0602]

451. **1990** Methods of solution of the Schrödinger equations for a class of anharmonic and singular potentials *Physics Letters A* [P-BARUT-W0409]

452. **1990** New frontiers in quantum electrodynamics and quantum optics *Proceedings, NATO Advanced Study Institute, Istanbul, Turkey, August 14-26, 1989* [P-BARUT-W0397]

453. **1990** Nonlinear de Broglie Waves and the Relation Between Relativistic and Non-Relativistic Solitons *Nonlinear Evolution Equations: Integrability and Spectral Methods* [P-BARUT-W0600]

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455. **1990** On the interpretation of the spinor phase change in a magnetic field in neutron interferometry *Physics Letters A* [P-BARUT-W0417]

456. **1990** On the measurement of the p anti-p total cross-section at s**(1/2) = 1.8-TeV *Hadronic Journal* [P-BARUT-W0398]

457. **1990** On the origin of mass, spin and wave particle duality *Conference proceedings, Teheran* [P-BARUT-W0399]

458. **1990** Pair production in an electric field in a time-dependent gauge *Physical Review D* [P-BARUT-W0404]

459. **1990** Path integral treatment of the hydrogen atom in a curved space of constant curvature. II. Hyperbolic space *Journal of Physics A: Mathematical and General* [P-BARUT-W0410]

460. **1990** Prediction of new tightly-bound states of H2+ (D2+) and 'cold fusion' experiments *International Journal of Hydrogen Energy* [P-BARUT-W0400]

461. **1990** Quantum electrodynamics based on self-fields: On the origin of thermal radiation detected by an accelerating observer *Physical Review A* [P-BARUT-W0406]

462. **1990** Quantum particle-like configurations of the electromagnetic field *Foundations of Physics Letters* [P-BARUT-W0415]

463. **1990** Quantum theory of single events: Localized De Broglie wavelets, Schrödinger waves, and classical trajectories *Foundations of Physics* [P-BARUT-W0418]

464. **1990** Reflection, transmission and spin rotation of dirac neutrinos in magnetic fields *Zeitschrift für Physik A Atomic Nuclei* [P-BARUT-W0419]

465. **1990** Regularized analytic evaluation of vacuum polarization in a Coulomb field *Physical Review D* [P-BARUT-W0413]

466. **1990** Renormalization and elimination of preacceleration and runaway solutions of the Lorentz-Dirac equation *Physics Letters A* [P-BARUT-W0411]

467. **1990** Self-field quantum electrodynamics: The two-level atom *Physical Review A* [P-BARUT-W0407]

468. **1990** Spin and polarization dynamics in nuclear and particle physics *Proceedings, Conference, Trieste, Italy, January 12-15, 1988* [P-BARUT-W0401]

469. **1990** Stability of the static solitons in a pure spinor theory with fractional power nonlinearities *International Journal of Modern Physics A* [P-BARUT-W0402]

470. **1990** The classical relativistic two-body problem with spin and self-interactions *Journal of Physics A: Mathematical and General* [P-BARUT-W0412]

471. **1990** The electron-positron system at short distances *Zeitschrift für Physik A Atomic Nuclei* [P-BARUT-W0416]

472. **1990** The remarkable algebra so*(2n), its representations, its Clifford algebra and potential applications *Journal of Physics A: Mathematical and General* [P-BARUT-W0408]

473. **1991** Brief History and Recent Developments in Electron Theory and Quantumelectrodynamics *The Electron* [P-BARUT-W0420]

474. **1991** Contribution of the individual discrete levels to the Lamb shift in hydrogen atoms *Journal of Physics B: Atomic, Molecular and Optical Physics* [P-BARUT-W0433]

475. **1991** Dynamical Symmetries of Relativistic Two-and Many Body Systems *Symmetries in Science V* [P-BARUT-W0421]

476. **1991** Exact Solutions of Spin (Rabi) Precession, Transmission and Reflection in the Eckhart Potential *Europhysics Letters (EPL)* [P-BARUT-W0428]

477. **1991** Exact solutions of the Dirac equation in spatially nonflat Robertson-Walker space-times. 2. [P-BARUT-W0422]

478. **1991** Internal dynamics of Majorana and Dirac infinite component wave equations and magnetic bound states *Goslar 1991, Classical and quantum systems* [P-BARUT-W0423]

1926 GENERATION EVIDENCE ATLAS
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479. **1991** Interpretation of self-field quantum electrodynamics *Physical Review A* [P-BARUT-W0431]

480. **1991** On Nonlinear Equations Admitting Soliton-Solutions with a de Broglie Phase *Communications in Theoretical Physics* [P-BARUT-W0604]

481. **1991** On the Mathematical Procedures of Selffield Quantumelectrodynamics *The Electron* [P-BARUT-W0424]

482. **1991** Path integral quantization of the magnetic top *Physics Letters A* [P-BARUT-W0434]

483. **1991** Polarization asymmetries in fermion-(anti)fermion scatterings with Dirac, Pauli, scalar, tensor and axial coupling factors *International Journal of Modern Physics A* [P-BARUT-W0435]

484. **1991** QED based on self-energy: The relativistic $2S_{1/2} \rightarrow 1S_{1/2}+1\gamma$ decay rates of hydrogenlike atoms *Physical Review A* [P-BARUT-W0429]

485. **1991** Quantum Theory of Single Events *Symposium on Foundations of Modern Physics 1990; ayrıca LNP 379* [P-BARUT-W0603]

486. **1991** Reflecting Slow Atoms from a Micromaser Field *Europhysics Letters (EPL)* [P-BARUT-W0427]

487. **1991** Selected Popular Writings of E.U. Condon [P-BARUT-W0425]

488. **1991** Selected Scientific Papers of E.U. Condon [P-BARUT-W0426]

489. **1991** The Relation Between Relativistic and Nonrelativistic Solitons and Kinks *Communications in Theoretical Physics* [P-BARUT-W0605]

490. **1991** The covariant many-body problem in quantumelectrodynamics *Journal of Mathematical Physics* [P-BARUT-W0432]

491. **1991** The einstein a-coefficient of spontaneous emission: a relativistic calculation in the Heisenberg representation *Zeitschrift für Physik D Atoms, Molecules and Clusters* [P-BARUT-W0430]

492. **1992** A covariant canonical decomposition of the bosonic and spinning extended objects with electric charge *Modern Physics Letters A* [P-BARUT-W0444]

493. **1992** Band structure for neutral magnetic dipoles in a periodic magnetic field: A simple spin polarizer *Physical Review Letters* [P-BARUT-W0445]

494. **1992** Construction of Pure Spinors and Neutrinos from the Electromagnetic Field *Conference C91-10-20.1 proceedings* [P-BARUT-W0436]

495. **1992** Effective coupling functions extracted from the scattering experiments with polarized protons at moderate energies *Conference C92-01-07 proceedings* [P-BARUT-W0437]

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497. **1992** Generalization of the Volkov Solution for Relativistic Particles with Anomalous Magnetic Moments *Europhysics Letters (EPL)* [P-BARUT-W0440]

498. **1992** How to avoid "quantum paradoxes" *Foundations of Physics* [P-BARUT-W0441]

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500. **1992** More on the radiation reaction and runaway solutions in classical electrodynamics *Physics Letters A* [P-BARUT-W0448]

501. **1992** Nonlinear nonlocal classical field theory of quantum phenomena *International Journal of Engineering Science* [P-BARUT-W0450]

502. **1992** On some mathematical problems in the evaluation of fields produced by currents which are generalized functions *Journal of Mathematical Physics* [P-BARUT-W0446]

503. **1992** On the optimum set of polarization measurements for the scattering of two identical fermions to determine the helicity amplitudes *Zeitschrift für Physik C Particles and Fields* [P-BARUT-W0454]

504. **1992** Particle-like configurations of the electromagnetic field: An extension of de Broglie's ideas *Foundations of Physics* [P-BARUT-W0451]

505. **1992** Path integrals and quantum interference *American Journal of Physics* [P-BARUT-W0452]

506. **1992** Relativistic theory of the Lamb shift in self-field quantum electrodynamics *Physical Review A* [P-BARUT-W0447]

507. **1992** Retarded radiation field and spontaneous emission for the hydrogen atom as an emitting antenna *Physical Review A* [P-BARUT-W0443]

508. **1992** The embedding of supersymmetry into the dynamical groups *AIP Conference Proceedings* [P-BARUT-W0438]

509. **1992** The magnetic top as a model of quantum spin *Annals of Physics* [P-BARUT-W0442]

510. **1992** Theories of extended objects and composite models of particles *Karuizawa 1992 proceedings* [P-BARUT-W0439]

511. **1992** There Are Two Quantum Theories — Not One *Bell's Theorem and the Foundations of Modern Physics* [P-BARUT-W0606]

512. **1993** A fundamental chirality in nature: The helical motion of the spinning electron *Conference C93-06-28.4 proceedings* [P-BARUT-W0455]

513. **1993** Classical relativistic spinning particle with anomalous magnetic moment: the precession of spin *Journal of Physics A: Mathematical and General* [P-BARUT-W0474]

514. **1993** Coherent states versus de Broglie wavelets *Conference C93-06-12.1 proceedings* [P-BARUT-W0456]

515. **1993** Derivation of quantum Maxwell equations from relativistic particle Hamiltonian *International Journal of Theoretical Physics* [P-BARUT-W0468]

516. **1993** Deterministic Wave Mechanics. A Bridge Between Classical Mechanics and Probabilistic Quantum Theory *The Interpretation of Quantum Theory: Where We Stand?* [P-BARUT-W0610]

517. **1993** Diffraction and interference of single De Broglie wavelets. Deterministic wave mechanics [P-BARUT-W0457]

518. **1993** Dirac's shell model of the electron and the general theory of moving relativistic charged membrabes *Physics Letters B* [P-BARUT-W0467]

519. **1993** Dynamical groups and supersymmetry II: Supersymmetric Aharonov-Bohm and anyon systems *Modern Physics Letters A* [P-BARUT-W0477]

520. **1993** Elastic e-p Scattering with Pauli Coupling *Turkish Journal of Physics* [P-BARUT-W0613]

521. **1993** Exact strongly bound solutions of the classical relativistic two-body problem due to magnetic interactions *Physica A: Statistical Mechanics and its Applications* [P-BARUT-W0470]

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523. **1993** Magnetic Top: On the Origin of Quantum Spin and Magnetic Moment *Classical and Quantum Systems — Foundations and Symmetries* [P-BARUT-W0609]

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1926 GENERATION EVIDENCE ATLAS
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525. **1993** New approach to the determination of eigenvalues and eigenfunctions for a relativistic two-fermion equation *Journal of Mathematical Physics* [P-BARUT-W0469]

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527. **1993** On Poisson brackets and symplectic structures for the classical and quantum zitterbewegung *Foundations of Physics* [P-BARUT-W0475]

528. **1993** On parity conservation and the question of the 'missing' (right-handed) neutrino *Modern Physics Letters A* [P-BARUT-W0465]

529. **1993** On the dynamical group of the system of two anyons with Coulomb interaction *Physics Letters A* [P-BARUT-W0462]

530. **1993** On the «new constants of the motion of the free Dirac electron» *Il Nuovo Cimento A* [P-BARUT-W0466]

531. **1993** Origin of Spin: From Rotation Group to Noncommutative Zitterbewegung *Symmetries in Science VI* [P-BARUT-W0458]

532. **1993** Polarization dynamics in nuclear and particle physics *Proceedings, 2nd Adriatico Research Conference, Trieste, Italy, January 7-10, 1992* [P-BARUT-W0459]

533. **1993** Radiation reaction for the classical relativistic spinning particle in scalar, tensor and linearized gravitational fields *Physics Letters B* [P-BARUT-W0471]

534. **1993** Space reflections in the dichotomic magnetic charge model *Journal of Physics A: Mathematical and General* [P-BARUT-W0463]

535. **1993** Spinor Solutions in Robertson Walker Type Curved Spaces *Communications in Theoretical Physics* [P-BARUT-W0608]

536. **1993** Spinors, the Dirac formalism and correct complex conjugation *International Journal of Modern Physics A* [P-BARUT-W0473]

537. **1993** Suggestions for Unifying Two Types of Quantized Redshift of Astronomical Bodies *Apeiron* [P-BARUT-W0612]

538. **1993** The Cartan formalism in field theory *Helvetica Physica Acta* [P-BARUT-W0460]

539. **1993** The Extensions of space-time. Physics in the eight-dimensional homogeneous space d = SU(2,2)/K *Conference C93-01-04.1 proceedings* [P-BARUT-W0461]

540. **1993** The Lamb shift in the charge-magnetic monopole system *Modern Physics Letters A* [P-BARUT-W0476]

541. **1993** «Localized» tachyonic wavelet solutions of the wave equation *Physics Letters A* [P-BARUT-W0472]

542. **1994** Accelerating Wavelets *Kaynakta "F. of Physics"* [P-BARUT-W0644]

543. **1994** Algebraic Perturbation Theory for Singular Potentials *Physics Letters* [P-BARUT-W0627]

544. **1994** An Update on Self-Field Quantum Electrodynamics *Plenum NATO Series* [P-BARUT-W0643]

545. **1994** Classical Relativistic 2-Body Problem without Spin — [P-BARUT-W0622]

546. **1994** Conformal space-times: The Arenas of physics and cosmology *Foundations of Physics* [P-BARUT-W0479]

547. **1994** Debating the final theory *Foundations of Physics* [P-BARUT-W0639]

548. **1994** Dynamical Groups as an Independent Formulation of Relativistic Quantum Theory / Dynamical Groups and Infinite Component Wave Equations *Fundamental Theories of Physics* [P-BARUT-W0625]

549. **1994** Electrodynamics at Short Distances and Particle Physics *Q594-Minsk* [P-BARUT-W0641]

550. **1994** Electromagnetic Interactions and Particle Physics / Foundations of Electrodynamics and Particle Physics *Frontiers of Fundamental Physics* [P-BARUT-W0633]

551. **1994** Enhancement of the rate of radiative processes in the field of a magnetic monopole *Physica Scripta* [P-BARUT-W0480]

552. **1994** From Infinite to Finite Number of Degrees of Freedom *Turkish Journal of Physics* [P-BARUT-W0636]

553. **1994** Further Results on the Theory of Stern-Gerlach Experiment — [P-BARUT-W0623]

554. **1994** Localized Rotating Spinning Wavelets of Spin-1/2 *Physics Letters A* [P-BARUT-W0635]

555. **1994** Localized Solutions of the Gravitational Field in Linear Approximation *Classical and Quantum Gravity* [P-BARUT-W0620]

556. **1994** Neutrinos from the Electromagnetic Field — [P-BARUT-W0632]

557. **1994** On Exact Spin Precession Equation in Relativistic Quantum and Classical Electron Theory. Connections to the BMT Equation *Reviews in Mathematical Physics* [P-BARUT-W0637]

558. **1994** On the Formulation of Electrodynamics from a Single Principle *Foundations of Physics* [P-BARUT-W0626]

559. **1994** On the Lie Group Analysis of Linear Systems of O.D.E. *Lie Groups and Their Applications* [P-BARUT-W0634]

560. **1994** On the Mathematical Description of the Soliton-Antisoliton Annihilation in a Josephson Transmission Line — [P-BARUT-W0624]

561. **1994** On the Zitterbewegung of the Relativistic Electron *European Journal of Physics* [P-BARUT-W0629]

562. **1994** Particle Models — [P-BARUT-W0640]

563. **1994** Path Integral Quantization of the Confined Solutions of the Relativistic 2-Body Problem *Physica A* [P-BARUT-W0628]

564. **1994** Quantization of Collective Regular Structures of Particles *Kaynakta "J. Appl. Physics Enhancement of the Rate of Radiative Processes"* [P-BARUT-W0616]

565. **1994** Quantization of Constraint Solutions *Physical Review A* [P-BARUT-W0615]

566. **1994** Radiation reaction and the electromagnetic energy momentum of moving relativistic charged membranes *Physics Letters B* [P-BARUT-W0481]

567. **1994** Relativistic two-body system in (1+1)-dimensional QED. 1. On the circle S(1) *Annals of Physics* [P-BARUT-W0482]

568. **1994** Return to the Deterministic Interpretation of Quantum Mechanics *Venice Conference on Schrödinger* [P-BARUT-W0618]

569. **1994** Semiclassical Quantization of the Magnetic Top — [P-BARUT-W0631]

570. **1994** Solution of the Retardation Problem in Relativistic 2-Body Problem *Turkish Journal of Physics* [P-BARUT-W0638]

571. **1994** Space-time models from the electromagnetic field *Helvetica Physica Acta* [P-BARUT-W0483]

572. **1994** Strings and membranes in electron theory *Conference C94-06-06.2 proceedings* [P-BARUT-W0484]

573. **1994** Symmetries of elementary particles revisited *Conference C94-08-08.2 proceedings* [P-BARUT-W0485]

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574. **1994** Symmetry scattering for SU(2,2) and its applications *Journal of Mathematical Physics* [P-BARUT-W0486]

575. **1994** Symmetry scattering for SU(2,2) and its applications *Nuclear Physics A* [P-BARUT-W0487]

576. **1994** The Critical Turning Points in the Solutions of the Magnetic Top Equations of Motion — [P-BARUT-W0630]

577. **1994** The Electromagnetic Top — [P-BARUT-W0617]

578. **1994** The Energy-Momentum Theory of the Electromagnetic Field — [P-BARUT-W0646]

579. **1994** The Exponential map for the conformal group O(2,4) *Journal of Physics A* [P-BARUT-W0488]

580. **1994** The Exponential map for the unitary group SU(2,2) *Journal of Physics A* [P-BARUT-W0489]

581. **1994** The Fallacy of the Arguments Against Local Realism in Quantum Phenomena *Waves and Particles in Light and Matter* [P-BARUT-W0614]

582. **1994** Unstable States in Quantum Electrodynamics — [P-BARUT-W0642]

583. **1995** Dirac equation in spatially nonflat Robertson-Walker space-times. II *International Journal of Modern Physics D* [P-BARUT-W0490]

584. **1995** Fizikçinin Doğaya Bakışı ve Temel Bilimlerin Hayatımızdaki Yeri *Bilim ve Teknik, Şubat 1995* [P-BARUT-W0494]

585. **1995** Models of particle states *International Journal of Theoretical Physics* [P-BARUT-W0491]

586. **1995** Quantum Systems: New Trends and Methods (QS 94) *Proceedings, conference C94-05-23.1* [P-BARUT-W0492]

587. **1995** Self-field quantum electrodynamics without infinities. A new calculation of vacuum polarization *Physics Letters A* [P-BARUT-W0493]

588. **1998** Spinor field as elementary excitations of a system of scalar fields *Foundations of Physics* [P-BARUT-W0496]

589. **1998** The energy-momentum tensor for electromagnetic interactions *Foundations of Physics* [P-BARUT-W0497]

WRITINGS AND OTHER RECORDED WORKS

1950 Strong Reflection Principle for each Fermion [P-BARUT-W0002]

1952 Letter to Yakup Bey (Zürich, 4 May 1952) [P-BARUT-W0007]

1964 Perturbation Theory: Methods of Quantum Field Theory in Statistical Physics [P-BARUT-W0044]

1964 Physics: Introduction to Advanced Field Theory. G. Barton. Interscience (Wiley), New York, 1963. x + 163 pp. Illus. $6.50. [P-BARUT-W0045]

1964 S-Matrix Theory: Mandelstam Theory and Regge Poles [P-BARUT-W0046]

1965 The Many-Body Problem: Quantum Field Theory and the Many-Body Problem [P-BARUT-W0066]

1966 Festschrift for Weisskopf: Preludes in Theoretical Physics, In Honor of V. F. Weisskopf [P-BARUT-W0069]

1966 Group Theory in Physics: Symmetry Groups in Nuclear and Particle Physics [P-BARUT-W0070]

1966 Theoretical Physics: High Energy Physics [P-BARUT-W0073]

1970 Handwritten note on the evolution of physical laws (Colorado, 2 Kasım 197?) [P-BARUT-W0113]

1971 Introduction to de Sitter and conformal groups and their physical applications [P-BARUT-W0123]

1982 Yaşam Öyküm [P-BARUT-W0268]

1983 Paul Dirac on His Eightieth Birthday [P-BARUT-W0283]

1983 To Eugene Paul Wigner on His Eightieth Birthday [P-BARUT-W0289]

1987 Book review [P-BARUT-W0334]

1990 Letter from Asım Barut to 'İsmail' (4 Şubat 199?) [P-BARUT-W0395]

1990 Letter from Asım Barut to 'İsmail' (Aralık 20, 199?) [P-BARUT-W0396]

1996 A. O. Barut, contribution in 'Non-Compact Groups' (ed. Y. Chow), p. 1 [P-BARUT-W0495]

n.d. Fiziksel matematik ders notları (unfinished lecture notes on mathematical physics) [P-BARUT-W0501]

n.d. Research problems written on cards (unpublished collection; Barut planned to have it printed) [P-BARUT-W0508]

n.d. Unpublished paper on a new form of probability theory [P-BARUT-W0513]

n.d. Monographs in Theoretical Physics [P-BARUT-W0516]

n.d. Science and Developing Countries [P-BARUT-W0517]

1990 Louis de Broglie and the Single Quantum Particle [P-BARUT-W0518]

1999 Yaşam öyküsü / otobiyografik metin[5] [P-BARUT-W0519]

1956 Grundlagen der Elektronenphysik [P-BARUT-W0522]

1968 Does the Proton Have an Internal Structure [P-BARUT-W0544]

1979 Unified Theories? [P-BARUT-W0568]

1981 Maddenin Gerçek Yapıtaşları Nelerdir [P-BARUT-W0571]

1982 Laser Probes of the Micro- and Macro-Cosmos [P-BARUT-W0575]

1987 Essay Review – Quantum Mechanics [P-BARUT-W0590]

1988 Essay Review: E. Schrödinger's Collected Papers [P-BARUT-W0592]

1989 Alfred Landé [P-BARUT-W0597]

1992 Closing Address [P-BARUT-W0607]

1994 Heinrich Rudolph Hertz and the Unification of Mechanics and Electromagnetism [P-BARUT-W0619]

1994 Self Energies and Radiation of Electron-Positron System [P-BARUT-W0621]

1994 Rencontres in Trieste [P-BARUT-W0645]

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n.d. On the Nature of Waves and Particles: Of what is the world made? [P-THE-BARUT-0023]

1995 Ünlü fizikçimiz Asım Barut’un doğaya bakışı [P-THE-BARUT-0028]

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